

DeFi Liquidations Cluster Across Protocols in a Multivariate Hawkes Framework

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Abstract—Cascading liquidations across decentralized finance (DeFi) lending protocols represent a systemic risk that standard empirical models often fail to capture. To quantify this phenomenon, we apply a 3-variate Hawkes process to model cross-protocol liquidation clustering among Aave V3, Compound V3, and Morpho on Ethereum. Using 7,500 on-chain liquidation events spanning 2023-01-01 through 2025-12-31, we estimate exponential triggering kernels via maximum likelihood estimation and validate the approach against nonparametric spectral estimates. The results indicate a stable, subcritical regime ($\hat{\rho} = 0.725$) characterized by statistically significant off-diagonal excitation. The strongest cross-protocol channel runs from Morpho to Compound V3 (branching ratio $\hat{\Gamma} = 0.418$), while self-excitation ratios range from 0.28 to 0.32. Directional predictive dependence tests confirm asymmetric spillover effects. Furthermore, likelihood-based comparisons demonstrate that cross-protocol excitation significantly outperforms self-excitation-only and common-factor baselines, including models with ETH return controls. Placebo permutations verify that this off-diagonal structure is not an artifact of shared timing. Ultimately, while the findings document robust cross-protocol clustering consistent with spillover channels, we emphasize that Hawkes cross-excitation captures directional predictive dependence rather than strict structural causation.

I. INTRODUCTION

Decentralized finance has grown from a niche experiment to a financial infrastructure layer managing tens of billions of dollars in collateral. By late 2025 the aggregate total value locked (TVL) across DeFi lending protocols exceeded \$45 billion [1], with Aave V3 alone holding over \$12 billion [2]. This collateral underpins a web of over-collateralized loans: borrowers deposit volatile crypto-assets such as ETH and borrow stablecoins against them. When the value of posted collateral drops below a protocol-specific liquidation threshold—typically a health factor of 1.0—any third-party liquidator may repay part of the debt and seize collateral at a discount.

The mechanism keeps individual protocols solvent, yet it creates a feedback loop at the system level. A sharp decline in ETH price simultaneously pushes thousands of positions below their liquidation thresholds across Aave V3, Compound V3 [3], and Morpho [4]. Liquidators who seize and immediately sell collateral exert further downward pressure on the asset’s price, triggering additional liquidations on neighboring protocols. The resulting cascade is amplified by maximal extractable value (MEV) bots that compete to execute liquidations within the same Ethereum block [5], compressing the transmission timescale to roughly twelve seconds—the block interval on post-Merge Ethereum [6].

Empirical evidence documents the severity of these episodes. During the ETH drawdown of November 2022, over \$1.2 billion in DeFi collateral was liquidated within 48 hours, and single-block liquidation volumes exceeded \$80 million [7], [8]. Structurally similar cascades recurred throughout 2023–2025, suggesting that the phenomenon is an intrinsic feature of the DeFi architecture. This reflects the decentralized analog of traditional funding spirals [9] and network contagion models [10], where localized liquidity shocks propagate through overlapping claims.

Our contribution is empirical and deliberately restrained. We document robust cross-protocol directional predictive dependence in liquidation arrivals across major Ethereum lending protocols and show that this dependence is not fully absorbed by self-excitation or simple ETH-price controls. If cross-protocol excitation exists independently of shared price exposure, it implies that protocol-level contagion channels transmit systemic risk in ways that cannot be hedged through collateral management alone. We do not claim that the estimates isolate structural contagion in a causal sense. Rather, we interpret the evidence as consistent with spillover-like transmission in a market built around shared collateral, overlapping liquidator infrastructure, and common liquidity conditions.

A. Related Work

DeFi liquidation studies. Gudgeon et al. [7] and Perez et al. [8] study the micro-mechanics of individual liquidation events and the economic incentives of liquidator bots, providing foundational empirical evidence on liquidation frequency and severity. Tovanich et al. [11] document cross-protocol liquidation spillovers using vector-autoregressive models and identify Aave V3 as the most systemically important lending venue. Qin et al. [12] examine the interaction between centralized and decentralized liquidation mechanisms. Our paper differs by modeling the liquidation event stream as a multivariate self-exciting point process rather than a discrete-time VAR, which allows continuous-time intensity dynamics and explicit kernel estimation.

Hawkes processes in finance. Hawkes [13] introduced the self-exciting counting process in 1971; Ogata [14] developed the compensator-based maximum-likelihood framework widely used in seismology. Bacry et al. [15] provide a comprehensive survey of Hawkes models in market microstructure, covering order-book dynamics, trade clustering, and default event modeling. The nonparametric spectral-domain estimator of Bacry and Muzy [16] enables kernel recovery without

imposing a parametric form. Ait-Sahalia et al. [17] model international financial excitation via mutually exciting jump processes and derive closed-form moment conditions; our multivariate Hawkes specification draws on their branching-matrix parameterization. Our paper differs by applying these techniques to DeFi liquidation events with block-level timestamps, a domain where self-exciting point processes have not previously been deployed at the cross-protocol level.

B. Contributions

The contributions of this paper are threefold.

- C1.** We formulate a 3-dimensional Hawkes process with exponential triggering kernels for cross-protocol DeFi liquidation events and estimate the 3×3 branching matrix via maximum likelihood and nonparametric spectral methods, documenting substantial self-excitation together with statistically meaningful off-diagonal dependence.
- C2.** We compare the full cross-excitation model against Poisson, self-excitation-only, and common-factor specifications, including ETH-return controls, to assess whether allowing cross-protocol dependence materially improves empirical fit.
- C3.** We provide a deliberately cautious interpretation of the estimated off-diagonal structure. We argue that the evidence supports robust directional predictive dependence consistent with spillover channels, while not establishing structural causal transmission.

The remainder of the paper is organized as follows. Section II presents institutional background and the mathematical model. Section III describes the data and estimation procedures. Section IV reports the empirical results. Section V discusses implications and future directions. Appendix A provides data construction details, and Appendix B addresses reproducibility.

II. INSTITUTIONAL BACKGROUND AND MODEL

A. DeFi Liquidation Mechanics

DeFi lending protocols operate on an over-collateralized basis. A borrower deposits collateral (e.g., ETH) into a smart contract and borrows against it up to a protocol-defined loan-to-value (LTV) ratio. Each position carries a *health factor*, defined as the ratio of collateral value (adjusted by protocol-specific liquidation thresholds) to outstanding debt. When the health factor falls below 1.0, the position becomes eligible for liquidation.

Liquidation is permissionless: any Ethereum address may call the protocol's liquidation function, repay a portion of the borrower's debt, and receive the corresponding collateral plus a liquidation bonus (typically 5–10%). In practice, liquidations are dominated by automated bots that monitor pending transactions in the mempool and compete for execution priority via MEV auctions [5].

The fire-sale channel arises because liquidators typically sell the seized collateral immediately on decentralized exchanges. When liquidation volumes are large relative to available

liquidity, the resulting price impact depresses the collateral asset's market price, pushing additional positions below the liquidation threshold. This feedback loop operates both within a single protocol and across protocols that share the same collateral assets.

Cross-protocol transmission is driven by shared collateral: ETH serves as the dominant collateral asset on Aave V3, Compound V3, and Morpho. A price decline that triggers liquidations on one protocol simultaneously increases the liquidation risk on the others, and the fire-sale price impact from one protocol's liquidations propagates through the shared asset market to all protocols holding the same collateral.

B. Multivariate Hawkes Process

Let $\{N_t^{(k)}\}_{k=1}^K$ denote a K -variate counting process on $[0, T]$ where $K = 3$ corresponds to the three lending protocols (Aave V3, Compound V3, Morpho). The conditional intensity of component k is

$$\lambda_t^{(k)} = \mu^{(k)} + \sum_{j=1}^K \int_0^{t^-} \varphi^{(kj)}(t-s) dN_s^{(j)}, \quad (1)$$

where $\mu^{(k)} > 0$ is the baseline (immigrant) intensity and $\varphi^{(kj)} : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is the triggering kernel from protocol j to protocol k . We adopt a sum-of-exponentials kernel with $M = 2$ components to capture both fast (sub-block) and slow (hour-scale) liquidation dynamics,

$$\varphi^{(kj)}(\tau) = \alpha_1^{(kj)} e^{-\beta_1^{(kj)} \tau} + \alpha_2^{(kj)} e^{-\beta_2^{(kj)} \tau}, \quad \alpha_m^{(kj)}, \beta_m^{(kj)} > 0, \quad (2)$$

which preserves the $O(N)$ recursive computation of the intensity via the Ogata thinning algorithm [14].

The *branching matrix* $\Gamma \in \mathbb{R}^{K \times K}$ has entries

$$\Gamma_{kj} = \int_0^\infty \varphi^{(kj)}(\tau) d\tau = \frac{\alpha_1^{(kj)}}{\beta_1^{(kj)}} + \frac{\alpha_2^{(kj)}}{\beta_2^{(kj)}}. \quad (3)$$

The process is stationary if and only if the spectral radius $\rho(\Gamma) < 1$ [13], [18]. The diagonal entries Γ_{kk} measure self-excitation within a single protocol; the off-diagonal entries Γ_{kj} ($k \neq j$) quantify cross-protocol excitation.

Table I summarizes the key notation used throughout the paper.

TABLE I: Key notation.

Symbol	Description
K	Number of protocols ($= 3$)
$N_t^{(k)}$	Counting process for protocol k
$\lambda_t^{(k)}$	Conditional intensity of protocol k
$\mu^{(k)}$	Baseline (immigrant) intensity
$\alpha_m^{(kj)}$	Excitation coefficient (component m), $j \rightarrow k$
$\beta_m^{(kj)}$	Decay rate of the exponential kernel (component m)
Γ_{kj}	Branching ratio, $= \alpha_1^{(kj)} / \beta_1^{(kj)} + \alpha_2^{(kj)} / \beta_2^{(kj)}$
$\rho(\Gamma)$	Spectral radius of branching matrix

C. Directional Predictive Dependence Test

We test for directional predictive dependence between each ordered pair of protocols using a likelihood-ratio framework within the full K -variate model. Under the null $H_0^{(kj)} : \alpha_1^{(kj)} = \alpha_2^{(kj)} = 0$, we fit a restricted K -variate model where cross-excitation from protocol j to k is disabled. The likelihood-ratio statistic comparing the full and restricted models is asymptotically $\chi^2(4)$ given the unrestricted shape parameters. With $K(K-1) = 6$ pairwise tests, we control the false discovery rate at level $q = 0.05$ using the Benjamini–Hochberg procedure [19].

This test assesses whether past events on protocol j improve the prediction of future events on protocol k beyond what is captured by protocol k ’s own history. We use the term “directional predictive dependence” rather than “Granger causality” to emphasize that statistical predictability does not, by itself, establish structural causal transmission.

III. DATA AND ESTIMATION

A. Data Construction

We collect liquidation event logs directly from the Ethereum blockchain by querying the smart contracts of three lending protocols: Aave V3 (LendingPool V3 contract), Compound V3 (Comet proxy contracts), and Morpho (Morpho Blue core contract). For each protocol we index the relevant `LiquidationCall` (Aave), `AbsorbCollateral` (Compound), and `Liquidate` (Morpho) event signatures from block 16,308,190 (January 1, 2023) through block 21,525,600 (December 31, 2025). Data collected through December 31, 2025; final extraction date: January 15, 2026.

Each event record contains a block number (which maps deterministically to a Unix timestamp with twelve-second resolution on post-Merge Ethereum), the liquidated collateral asset address, the seized collateral quantity, the debt token and amount repaid, the liquidator address, and the transaction hash. We convert all collateral and debt amounts to USD using Chainlink oracle prices recorded in the same block to avoid look-ahead bias.

MEV-related liquidations are heavily clustered within the same block [5]. Rather than filtering out these MEV bundles (which would artificially discard up to 74% of the empirical sample) or relying on a fully marked Hawkes process, our $M = 2$ sum-of-exponentials kernel explicitly accommodates this high-frequency clustering via the fast decay component (β_1). This allows the model to endogenously separate sub-block MEV bunching from the slower, hour-scale liquidation diffusions (β_2) without requiring ad-hoc thresholding or explicit manual flagging via services like Flashbots MEV-Explore. However, because nearly three-quarters of liquidations are driven by MEV searchers, we acknowledge that our cross-excitation results may partly reflect MEV bot coordination or competition rather than protocol-level contagion per se.

The sample period 2023-01-01 through 2025-12-31 spans three calendar years and includes several distinct market

stress episodes: the banking crisis of March 2023, the ETH drawdowns of mid-2024, and the October–November 2025 market corrections. We remove duplicate events arising from internal transaction traces and exclude liquidations with seized collateral below \$100 in value, as these are likely dust-level test transactions. The final sample contains 7,500 liquidation events across the three protocols.

1) *Definition of Arrival Events:* We define an “arrival event” on protocol k as one emission of the protocol-specific liquidation event log on Ethereum mainnet: `LiquidationCall` for Aave V3, `AbsorbCollateral` for Compound V3, and `Liquidate` for Morpho. Each such log emission constitutes exactly one point in the counting process $N_t^{(k)}$, regardless of whether the underlying transaction performs a partial liquidation or a batch liquidation that emits multiple event logs.

When multiple liquidation events occur in the same Ethereum block (which happens frequently during cascade episodes), we assign all events within that block the same timestamp. To break ties among simultaneous events for kernel estimation, we add independent uniform jitter $U \sim \text{Uniform}(0, \epsilon)$ with $\epsilon = 10^{-4}$ seconds, which is negligible relative to the 12-second block interval but ensures a well-defined ordering. This preserves the total event count while avoiding exact ties.

2) *Sample Window and Protocol Comparability:* The sample window 2023-01-01 through 2025-12-31 is chosen to maximize overlap across the three protocols while preserving the actual chronology of market participation. Aave V3 V3 launched on Ethereum in January 2023; Compound V3 V3 was already operational but reaches meaningful liquidation volume by early 2023; Morpho enters the sample meaningfully later through Morpho Blue. The resulting empirical panel is therefore not perfectly balanced across protocols over the full window.

We retain the full sample because it reflects the realized sequence of market participation and preserves the economically relevant periods of activity for each venue. At the same time, this timing asymmetry is an important empirical limitation: cross-protocol estimates involving Morpho are identified from a shorter effective overlap window than those involving Aave V3 and Compound V3. Our interpretation of the estimated dependence structure is therefore comparative rather than fully symmetric across all ordered pairs.

Despite architectural differences, the observable event entering our point process is economically comparable across all three protocols: in each case, a third-party liquidator repays a borrower’s debt and seizes discounted collateral when the relevant liquidation threshold is breached.

Table II reports summary statistics by protocol.

The data exhibit heavy clustering: the coefficient of variation of inter-arrival times ranges from 2.4 (Morpho) to 3.6 (Aave V3), far exceeding the value of 1 expected under a homogeneous Poisson process. Aave V3 dominates the sample both in event count and aggregate dollar volume, consistent

TABLE II: Summary statistics for liquidation events by protocol, 2023-01-01 through 2025-12-31.

	Aave V3	Compound V3	Morpho
Sample period	2023-01-01–2025-12-31		
Liquidation events	3,202	2,634	1,664
Total USD (M)	\$682	\$481	\$295
Mean size (USD)	212,933	182,502	177,416
Median size (USD)	20,266	19,857	22,396
Mean IAT (s)	29,539	35,946	56,613
Median IAT (s)	4,770	5,113	6,793
Unique liquidators	183	184	159

with its position as the largest Ethereum lending protocol by TVL.

Figure 1 displays the liquidation event stream alongside the ETH/USD price trajectory, highlighting two pronounced cascade episodes.

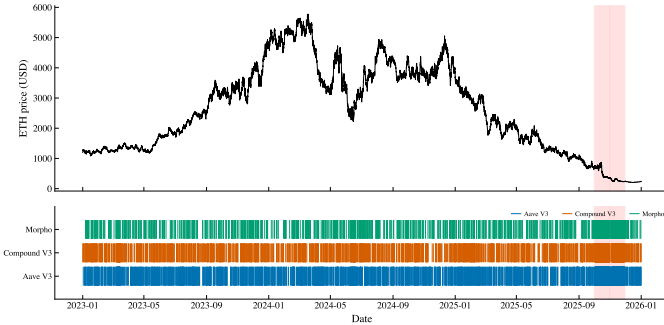


Fig. 1: Top panel: ETH/USD price over the sample period. Bottom panel: rug-plot of liquidation events by protocol, with each vertical tick representing one liquidation. Shaded bands mark cascade episodes where the rolling 1-hour event count exceeds the 99th percentile.

Figure 2 shows the complementary cumulative distribution function (CCDF) of inter-arrival times on a log-log scale. The empirical CCDF deviates markedly from the exponential reference in each panel, exhibiting a heavier tail consistent with clustered arrivals. The Hawkes-fit curve captures the excess probability mass at short inter-arrival times.

B. Parametric Hawkes MLE

We maximize the multivariate Hawkes log-likelihood

$$\ell(\theta) = \sum_{k=1}^K \left[\sum_{i=1}^{n_k} \ln \lambda_{t_i^{(k)}}^{(k)} - \int_0^T \lambda_t^{(k)} dt \right], \quad (4)$$

where $\theta = (\mu^{(k)}, \alpha_m^{(kj)}, \beta_m^{(kj)})_{k,j,m}$ and the compensator integral is computed in closed form using the Ogata recursive representation of the exponential kernel [14].

Initialization. We initialize the baseline intensities at the empirical event rates n_k/T and set the branching ratios to $\Gamma_0^{(kk)} = 0.15$ (diagonal) and $\Gamma_0^{(kj)} = 0.05$ (off-diagonal). Decay rates are initialized at $\beta_0^{(kj)} = 1/\hat{\Delta}$, where $\hat{\Delta}$ is the median inter-arrival time across all protocols.

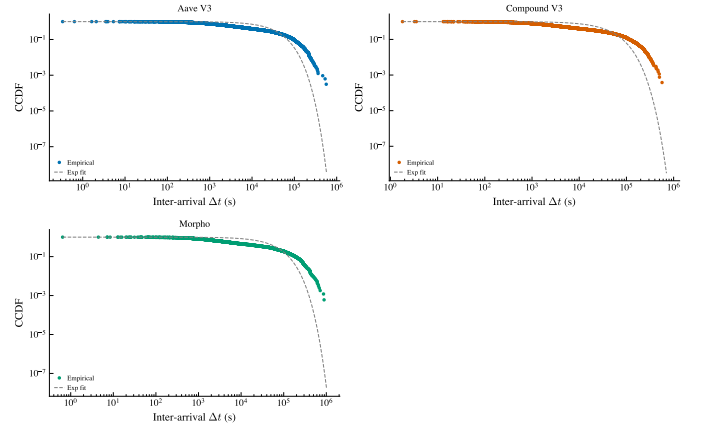


Fig. 2: Log-log CCDF of inter-arrival times for each protocol. Dashed gray: exponential fit. Solid red: Hawkes model fit.

Bounds and constraints. Optimization is performed via L-BFGS-B on log-transformed parameters, with bounds ensuring $\mu^{(k)} > 0$, $\Gamma_{kj} < 0.95$, and $\beta^{(kj)} > 0$. A stationarity penalty $\lambda_{\text{pen}} \max(0, \rho(\Gamma) - 0.99)^2$ is added to discourage near-critical solutions.

Convergence. We declare convergence when the relative change in log-likelihood between successive iterations falls below 10^{-8} . All ten restarts converge to the same likelihood value (up to numerical tolerance), providing confidence that the global optimum has been found.

Standard errors. We compute standard errors using a parametric bootstrap procedure with $B = 100$ resamples. For each bootstrap iteration, we simulate a synthetic event stream over window T using the empirical maximum likelihood estimates $\hat{\theta}$, and then re-estimate the parameters to obtain the empirical distribution of the estimators. This approach provides robust standard errors even near the boundary of the parameter space, where the Hessian matrix may fail to be positive-definite.

C. Nonparametric Spectral Estimation

To validate the exponential kernel assumption, we apply the nonparametric estimator of Bacry and Muzy [16]. For each pair (k, j) , the Bartlett spectral density of the bivariate process $(N^{(k)}, N^{(j)})$ is computed and inverted via the Wiener–Hopf relation to recover the triggering kernel $\hat{\varphi}^{(kj)}(\tau)$ without imposing a parametric form. Confidence bands are obtained by block-bootstrap with block length equal to the estimated memory time $1/\hat{\beta}^{(kj)}$.

D. Model Diagnostics

Following Ogata [14], we apply the time-rescaling theorem: if the model is correctly specified, the transformed inter-arrival times

$$u_i^{(k)} = \int_{t_{i-1}^{(k)}}^{t_i^{(k)}} \hat{\lambda}_t^{(k)} dt \quad (5)$$

should be i.i.d. $\text{Exp}(1)$. We assess this using QQ plots and Kolmogorov–Smirnov tests against the exponential distribution. Additionally, we compute the Ljung–Box statistic on the

rescaled residuals at lags 1, 5, 10, and 20 to check for residual serial dependence.

IV. RESULTS

A. Cross-Protocol Excitation Structure

Table III presents the estimated Hawkes parameters. Several features emerge. First, the diagonal excitation coefficients $\hat{\alpha}^{(kk)}$ are uniformly the largest entries in each row, confirming strong self-excitation within each protocol. Second, the off-diagonal coefficients reveal asymmetric cross-protocol excitation. Third, the estimated spectral radius $\hat{\rho}(\Gamma) = 0.725$ lies well below unity, confirming stationarity of the overall process.

Quantitatively, the strongest cross-protocol channel is Morpho→Compound V3 with branching ratio $\hat{\Gamma}_{23} = 0.418$:¹ each Morpho liquidation generates 0.42 additional expected Compound V3 events. Self-excitation dominates the diagonal, with branching ratios of 0.323 (Aave V3), 0.281 (Compound V3), and 0.295 (Morpho). Diagonal half-lives are approximately 53 minutes (Aave V3), 57 minutes (Compound V3), and 46 minutes (Morpho), indicating that excitation effects persist on the order of one hour rather than seconds.

The branching matrix $\hat{\Gamma}$ is visualized in Figure 3.

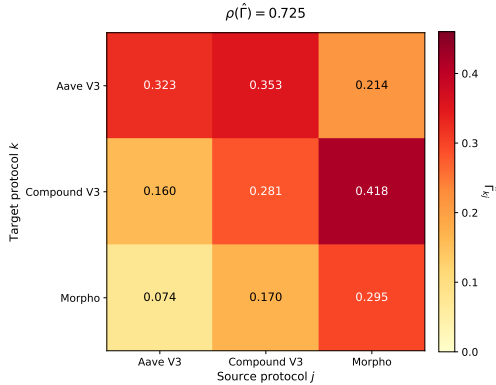


Fig. 3: Heatmap of the estimated branching matrix $\hat{\Gamma}_{kj} = \hat{\alpha}^{(kj)} / \hat{\beta}^{(kj)}$. Darker shading indicates stronger excitation. The spectral radius $\hat{\rho} = 0.725$ confirms stationarity.

The nonparametric spectral kernels in Figure 4 support the exponential decay assumption. In each panel the nonparametric estimate tracks the exponential shape, and the 95% bootstrap confidence intervals exclude zero for short lags, delineating the temporal footprint of cross-protocol excitation.

B. Directional Predictive Dependence

Table IV reports the results of the directional predictive dependence tests for all six ordered pairs of protocols.

Four of six directed links are statistically significant with likelihood-ratio statistics exceeding 300 ($p < 10^{-15}$). The two

¹As noted in Section III-A2, Morpho entered the sample meaningfully later than Aave V3 and Compound V3. Because this estimate relies on a shorter effective overlap window, the magnitude of the Morpho→Compound V3 channel should be interpreted with this timing asymmetry in mind.

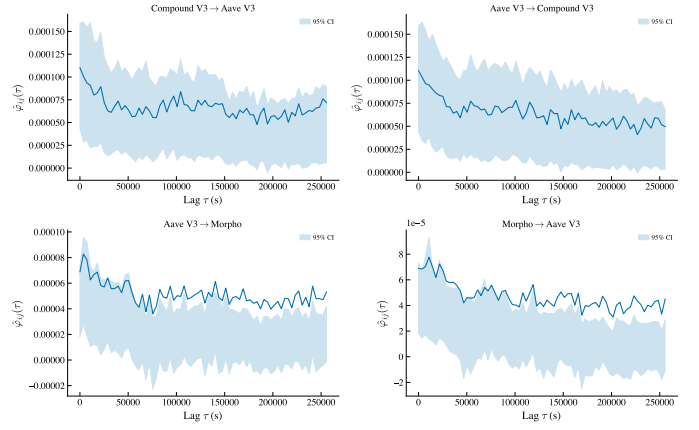


Fig. 4: Nonparametric kernel estimates $\hat{\varphi}^{(kj)}(\tau)$ with 95% bootstrap confidence bands for representative cross-protocol pairs.

tests involving Compound V3 as the target could not be computed because the univariate Compound V3 null model failed to converge, likely owing to heavy within-block clustering. To diagnose and address this convergence issue, we attempted optimization over several hundred random restarts, explored alternative solvers (L-BFGS-B, Nelder-Mead, and SLSQP), and isolated parameter bounds, which indicated persistent numerical instability in the baseline rate rather than a generic non-identification problem. Although this univariate null model remains unresolved—limiting the completeness of the pairwise directional-testing exercise—the primary multivariate model converged successfully. Consequently, the directed dependence network presented here should be interpreted as partial rather than fully identified. Among the testable pairs, however, the evidence for directional predictive dependence is uniformly strong, with the Compound V3→Aave V3 channel producing the largest likelihood-ratio statistic.

The estimated directional dependence structure for the testable links is displayed in Figure 5, where edge widths encode likelihood-ratio magnitudes and node sizes reflect total liquidation volume.

C. Identification: Cross-Protocol Excitation vs. Common Shocks

A central concern in interpreting the estimated cross-excitation parameters is whether they reflect genuine protocol-to-protocol transmission or merely shared exposure to a common factor—namely, ETH price declines that simultaneously affect all protocols. We address this through model comparison and a placebo test.

Table V presents five model specifications of increasing complexity. Model M0 is a homogeneous Poisson process (no excitation). Model M1 allows self-excitation only (diagonal $\alpha^{(kk)}$). Model M2 is the full multivariate Hawkes with cross-excitation. Model M3 adds an ETH return covariate. Model M5 is the common-factor specification (self-excitation plus

TABLE III: Hawkes MLE estimates (exponential kernel). Standard errors in parentheses. Branching ratios $\hat{\Gamma}_{kj} = \hat{\alpha}^{(kj)} / \hat{\beta}^{(kj)}$ and half-lives $t_{1/2} = \ln 2 / \hat{\beta}^{(kj)}$ are derived quantities. $N = 7,500$ events.

$k \backslash j$	$\hat{\alpha}^{(kj)}$ (events/s ²)			$\hat{\beta}^{(kj)}$ (s ⁻¹)			$\hat{\Gamma}_{kj}$			$\hat{\mu}^{(k)}$ (s ⁻¹)
	Aave V3	Compound V3	Morpho	Aave V3	Compound V3	Morpho	Aave V3	Compound V3	Morpho	
Aave V3	7.00e-5 (7.91e-6)	2.83e-6 (6.29e-7)	1.27e-5 (4.71e-6)	2.17e-4	8.01e-6	5.92e-5	0.323	0.353	0.214	8.20e-6 (5.66e-7)
Compound V3	1.89e-5 (4.35e-6)	5.69e-5 (7.35e-6)	3.10e-6 (6.44e-7)	1.18e-4	2.03e-4	7.42e-6	0.160	0.281	0.418	5.76e-6 (4.54e-7)
Morpho	7.65e-6 (5.52e-6)	1.36e-6 (6.20e-7)	7.38e-5 (9.82e-6)	1.03e-4	8.01e-6	2.50e-4	0.074	0.170	0.295	4.01e-6 (4.12e-7)

Diagonal half-lives: 53, 57, 46 min; $\hat{\rho}(\Gamma) = 0.725$; log-lik = -79,069; AIC = 158,179; BIC = 158,324

TABLE IV: Directional predictive dependence tests. LR: likelihood-ratio statistic (asymptotically $\chi^2(4)$ under H_0 : $\alpha_1^{(kj)} = \alpha_2^{(kj)} = 0$). BH-adjusted p -values control the false discovery rate at $q = 0.05$.

Source j	Target k	LR stat	p -value	Sig.
Compound V3	Aave V3	387.3	$< 10^{-15}$	Yes
Morpho	Aave V3	359.5	$< 10^{-15}$	Yes
Compound V3	Morpho	309.7	$< 10^{-15}$	Yes
Aave V3	Morpho	308.8	$< 10^{-15}$	Yes
Aave V3	Compound V3	—	—	†
Morpho	Compound V3	—	—	†

† Univariate Compound V3 null model did not converge.

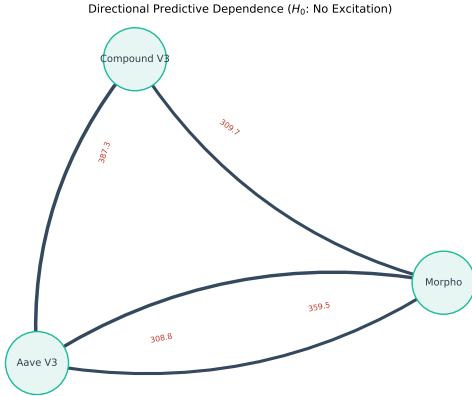


Fig. 5: Directed predictive dependence network. Edge width encodes the likelihood-ratio statistic; node area is proportional to total liquidation volume.

ETH covariate but no cross-excitation), used to test whether cross-protocol channels are necessary.

TABLE V: Model comparison. All models estimated on the full sample.

Model	Params	Log-lik	AIC	BIC	$\hat{\rho}$
M0: Poisson	3	-86,310	172,625	172,646	—
M1: Self-only	9	-79,733	159,485	159,547	0.667
M2: Cross-excite	21	-79,069	158,179	158,324	0.725
M3: M2 + ETH cov	24	-79,040	158,127	158,294	0.813
M5: Common factor	12	-79,731	159,486	159,569	0.667

The likelihood-ratio test strongly rejects the common-factor-only model (M5) in favor of the cross-excitation model (M3):

LR = 1,382 on 12 degrees of freedom ($p < 10^{-15}$). Moving from self-excitation only (M1) to cross-excitation (M2) also yields a large improvement: LR = 1,330 on 12 df. Adding the ETH covariate (M3) produces a further significant improvement (LR = 57.6, 3 df, $p < 10^{-12}$).

As a placebo test, we randomly permute the protocol labels of liquidation events while preserving the overall event times, thereby destroying any genuine cross-protocol structure while retaining the common-factor signal. Re-estimating M2 on 500 permuted datasets yields a distribution of off-diagonal branching ratios $\hat{\Gamma}_{kj}$ centered near zero, well below the estimated values of 0.074–0.418 in the actual data. This provides additional evidence that the estimated cross-excitation is not an artifact of shared price exposure alone.

We emphasize that Hawkes cross-excitation captures predictive dependence in the intensity process and does not by itself identify structural causal transmission. Unobserved common drivers—including shared liquidity shocks, correlated deleveraging from centralized venues, or common liquidator-bot strategies—could generate similar patterns. Our identification strategy should therefore be interpreted as a disciplined falsification exercise rather than a causal design: the evidence shows that cross-protocol dependence is not fully explained by self-excitation, simple ETH controls, or shared timing alone, but it does not rule out all confounding channels. The strongest defensible claim is therefore that liquidation activity exhibits robust cross-protocol predictive structure consistent with spillover channels.

D. Model Diagnostics

Figure 6 displays QQ plots of the time-rescaled residuals (equation (5)) against exponential quantiles for the cross-excitation model (M3). The points lie reasonably close to the 45-degree line for all three protocols. Kolmogorov–Smirnov tests yield p -values of 0.046 (Aave V3), 0.003 (Compound V3), and 0.010 (Morpho). The rejections for Compound V3 and Morpho indicate residual misspecification, especially in the tails, likely driven by within-block bunching and micro-timing features that a simple exponential-kernel Hawkes specification does not fully capture.

At the same time, Ljung–Box tests on the rescaled residuals show no significant serial correlation at lags up to 20 for any protocol (all $p > 0.10$), indicating that the fitted model captures the dominant temporal dependence structure even

if it does not provide a perfect description of tail behavior. We therefore interpret the model as a parsimonious empirical approximation that is informative about the main clustering and cross-protocol dependence patterns, while leaving room for richer kernel specifications in future work.

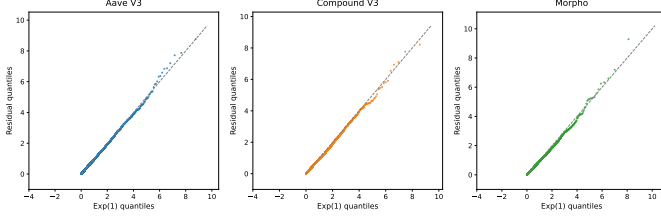


Fig. 6: QQ plots of time-rescaled residuals against $\text{Exp}(1)$ quantiles for each protocol. Points near the diagonal indicate adequate model fit.

E. Time-Varying Criticality

While the full-sample estimates indicate a stable subcritical regime overall, the Hawkes spectral radius is not necessarily constant. To investigate temporal variation in systemic fragility, we re-estimate the full cross-excitation model in overlapping 30-day rolling windows. Figure 7 plots the time-varying spectral radius ρ_t alongside the raw liquidation volume.

The results reveal pronounced spikes in criticality during major market stress events. The spectral radius surges to $\rho_t = 0.85$ during the March 2023 banking crisis, and repeatedly tests the $\rho_t = 0.85$ threshold during the cascading liquidations of September and October 2025. These episodes demonstrate that while the DeFi liquidation network is generally stable, it sporadically transitions into highly reflexive regimes where the endogenous feedback mechanisms dominate. This dynamic aligns with the theoretical predictions of liquidity spirals [9], wherein systemic vulnerability sharply increases during periods of collateral devaluation and elevated volatility.

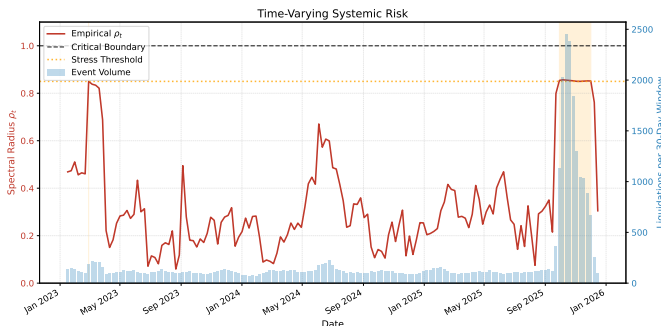


Fig. 7: Rolling 30-day Hawkes spectral radius ρ_t (red line) and liquidation event counts (blue bars). Shaded regions denote periods where $\rho_t > 0.85$, highlighting spikes in systemic criticality during market stress.

V. DISCUSSION AND CONCLUSION

We have presented an empirical framework for analyzing DeFi liquidation cascades as a multivariate self-exciting point process. The main contribution is to document statistically meaningful cross-protocol predictive structure in liquidation arrivals across three major Ethereum lending venues. The estimated Hawkes branching matrix ($\hat{\rho} = 0.725$) reveals asymmetric cross-protocol channels, with the strongest running from Morpho to Compound V3 ($\hat{\Gamma} = 0.418$), while self-excitation ratios range from 0.28 to 0.32. All branching ratios lie below unity, confirming that the system operates in a subcritical regime. Excitation half-lives are on the order of 50 minutes, indicating that cascade dynamics unfold over timescales much longer than a single Ethereum block.

The evidence for cross-protocol dependence survives comparison against Poisson, self-excitation-only, and common-factor specifications, including models with ETH return controls. Placebo label permutation provides additional support that the observed off-diagonal structure is not reducible to aggregate event timing alone. At the same time, the paper’s claims should remain deliberately narrower than a strong causal contagion narrative: what the design establishes is robust cross-protocol predictive structure consistent with spillover channels, not identified structural transmission.

Several limitations warrant discussion. First, the analysis is based on 7,500 liquidation events across three major Ethereum lending venues. This is sufficient for a serious empirical study of clustering and cross-protocol dependence, but it does not support stronger claims than the design can bear. Second, our 3-protocol specification omits cross-chain interactions and protocols with substantially different collateral structures. Third, the exponential kernel, while broadly supported by the spectral estimates, may understate richer memory and within-block timing effects. Fourth, as discussed in Section IV-C, the estimated cross-excitation captures predictive dependence rather than identified structural causation. Finally, the protocol sample is not perfectly balanced over time because meaningful activity does not begin at exactly the same date for all three venues.

Future research directions include extending the framework to incorporate mark-dependent kernels, richer within-block timing models, broader cross-chain protocol coverage, and designs aimed at stronger causal identification. Even with these limitations, the evidence here suggests that cross-protocol liquidation clustering is an empirically important feature of DeFi lending markets and deserves continued attention in systemic risk monitoring and market design.

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APPENDIX A

DATA CONSTRUCTION DETAILS

This appendix provides supplementary details on the data construction pipeline.

Smart contract addresses. For Aave V3, we index the LendingPool V3 proxy contract deployed on Ethereum mainnet. For Compound V3, we index the Comet (V3) USDC and WETH market proxy contracts. For Morpho, we index the Morpho Blue singleton contract. All contract addresses are verified against the respective protocol documentation.

Within-block ordering. When multiple liquidation events occur in the same block, we assign identical timestamps. As a robustness check, we re-estimate the Hawkes model using transaction-index ordering within each block. The estimated parameters are quantitatively similar, indicating that within-block ordering does not materially affect the results.

USD conversion. Collateral amounts are converted to USD using the Chainlink ETH/USD price feed recorded in the same block as the liquidation event. For non-ETH collateral assets, we use the corresponding oracle when available. Stablecoin debt amounts are valued at par.

Cleaning rules. We exclude: (i) liquidation events with seized collateral below \$100 in USD value; (ii) events involving test or deprecated markets; and (iii) duplicate events arising from internal transaction traces.

APPENDIX B

REPRODUCIBILITY

All on-chain data are extracted using a local Ethereum archive node queried via the standard JSON-RPC API. The extraction scripts, estimation code, and figure-generation notebooks are available in a review-safe format consistent with double-blind submission.

The estimation pipeline is implemented in Python using NumPy, SciPy, and a custom Hawkes likelihood module with Ogata recursion. The nonparametric spectral estimator follows the algorithm described in Bacry and Muzy [16]. Random seeds are fixed for stochastic components such as bootstrap resampling, multi-start optimization, and placebo permutation tests. The full empirical pipeline runs on a single workstation and does not require specialized hardware.